

Exploratory Data Analysis of Air Pollution in India

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Abstract—This project presents an exploratory data analysis (EDA) study on air pollution data collected from multiple cities in India. The objective of the project is to investigate relationships between air pollutants, identify patterns in pollution levels, and apply dimensionality reduction and clustering techniques to discover hidden structures within the dataset. The analysis includes data preprocessing, visualisation, feature engineering, principal component analysis (PCA), and K-Means clustering. The findings demonstrate strong correlations between particulate pollutants and reveal meaningful pollution clusters across cities and seasons.

I. INTRODUCTION

Air pollution is one of the most significant environmental and public health challenges worldwide. India experiences particularly severe air quality problems due to rapid urbanisation, industrial activities, vehicle emissions, and seasonal environmental factors.

This project investigates air quality data collected from Indian cities using exploratory data analysis (EDA) techniques. The objectives of this study are:

- Analyse pollutant distributions and relationships.
- Identify trends and seasonal patterns.
- Apply feature engineering to improve data representation.
- Use Principal Component Analysis (PCA) for dimensionality reduction.
- Apply clustering techniques to identify pollution groups.

The dataset used in this project was obtained from Kaggle and contains measurements of multiple pollutants including PM2.5, PM10, NO₂, SO₂, CO, and O₃.

II. DATA PROCESSING

The dataset initially contained missing values, duplicated rows, and inconsistent measurements. Data preprocessing was therefore necessary before performing analysis.

A. Missing Values

Missing values were identified using summary statistics and visual inspection. Median imputation was applied to numerical pollutant variables because pollution data is often skewed and affected by outliers.

	Missing Values	Percentage
Xylene	18109	61.322001
PM10	11140	37.723071
NH3	10328	34.973418
Toluene	8041	27.229014
Benzene	5623	19.041008
AQI	4681	15.851139
AQI_Bucket	4681	15.851139
PM2.5	4598	15.570079
NOx	4185	14.171549
O3	4022	13.619586
SO2	3854	13.050692
NO2	3585	12.139785
NO	3582	12.129626
CO	2059	6.972334
Date	0	0.000000
City	0	0.000000

Fig. 1. Percentage of missing values across dataset variables.

B. Data Cleaning

Duplicate rows were removed to improve data quality. The date column was converted into a datetime format to enable time-series analysis.

C. Feature Scaling

Standardisation was applied before PCA and clustering because pollutant variables use different numerical scales.

III. DATA PLOTTING AND EXPLORATORY ANALYSIS

Several visualisation techniques were used to investigate the dataset.

A. Distribution Analysis

Histograms showed that pollutant variables such as PM2.5 and PM10 indicate the presence of severe pollution events.

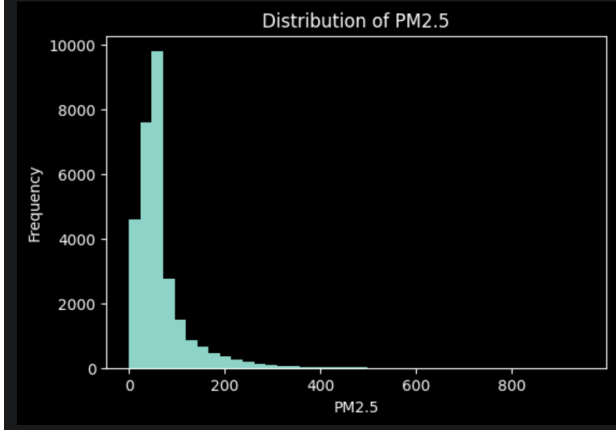


Fig. 2. Distribution of PM2.5 concentrations.

B. Correlation Analysis

A correlation heatmap revealed strong positive relationships between PM2.5 and PM10, suggesting that they may originate from similar pollution sources.

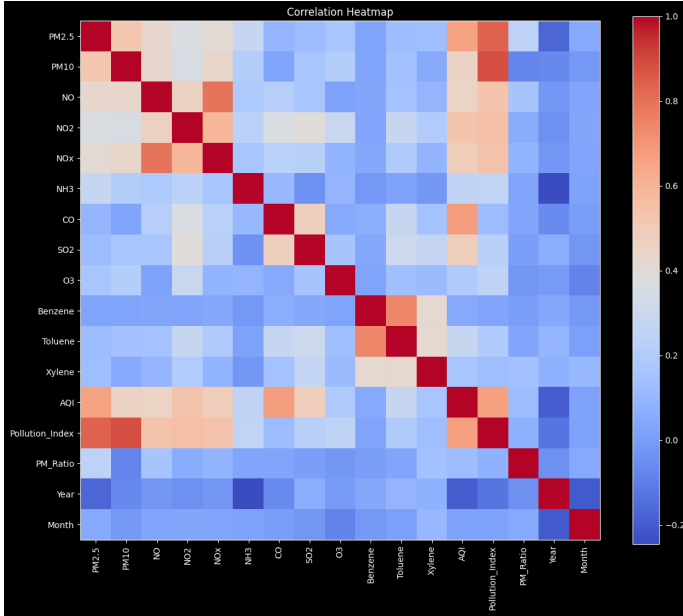


Fig. 3. Correlation heatmap of numerical pollutant variables.

C. Temporal Trends

Time-series analysis showed seasonal variation in pollution levels, with higher concentrations observed during colder months.

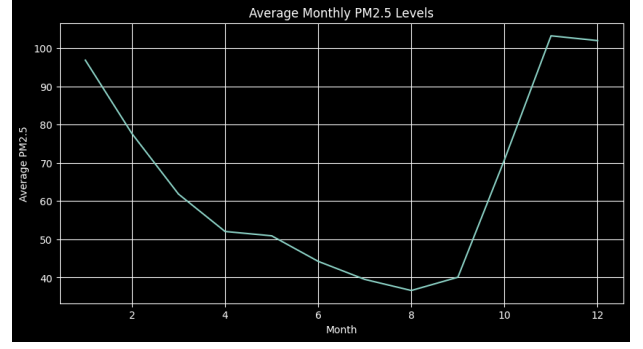


Fig. 4. Average monthly PM2.5 concentration levels.

D. City Comparisons

Comparisons between cities demonstrated substantial variation in pollution severity across India.

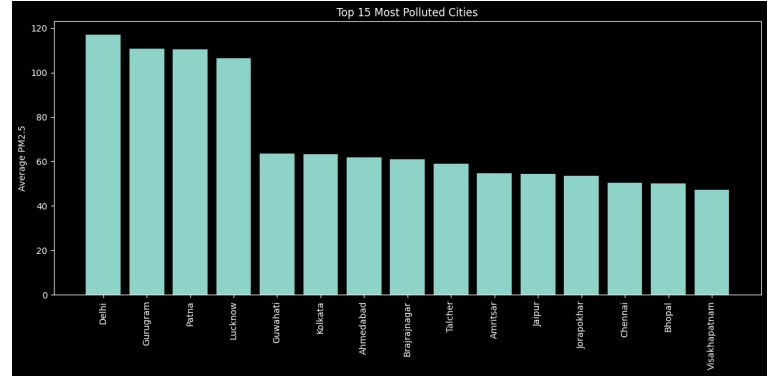


Fig. 5. Average PM2.5 levels across major Indian cities.

IV. FEATURE ENGINEERING

Feature engineering was applied to improve the descriptive power of the dataset.

A. Pollution Index

A combined pollution index was created using:

$$Pollution\ Index = \frac{PM2.5 + PM10 + NO_2}{3} \quad (1)$$

This feature provides a simplified representation of overall pollution severity.

B. PM Ratio

The PM2.5 to PM10 ratio was calculated to investigate the dominance of fine particulate matter.

$$PM\ Ratio = \frac{PM2.5}{PM10} \quad (2)$$

C. Temporal Features

Additional temporal features including month and year were extracted from the date column.

V. PRINCIPAL COMPONENT ANALYSIS

Principal Component Analysis (PCA) was applied to reduce dimensionality and identify dominant pollution patterns.

The explained variance analysis demonstrated that the first few principal components captured a large proportion of the dataset variance.

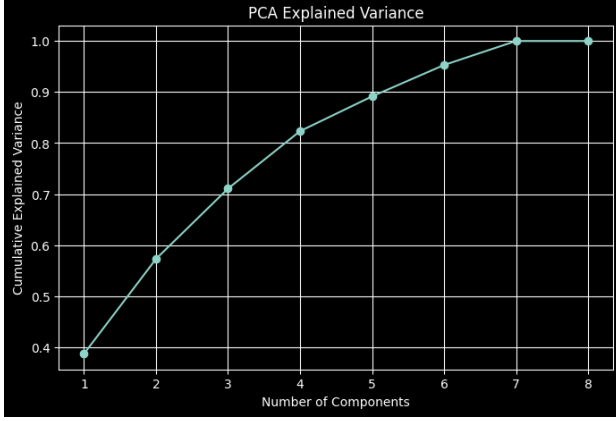


Fig. 6. Cumulative explained variance from PCA components.

The PCA projection revealed clear groupings within the data, indicating the presence of underlying pollution structures.

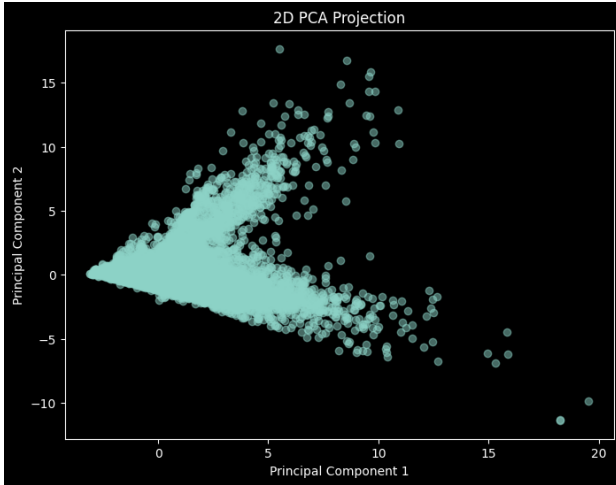


Fig. 7. 2D PCA projection of air pollution observations.

Feature loading analysis showed that PM_{2.5} and PM₁₀ contributed strongly to the first principal component.

VI. CLUSTERING

K-Means clustering was applied to group observations with similar pollution characteristics.

The Elbow Method was used to select the optimal number of clusters. Three clusters were chosen based on inertia reduction and cluster separation.

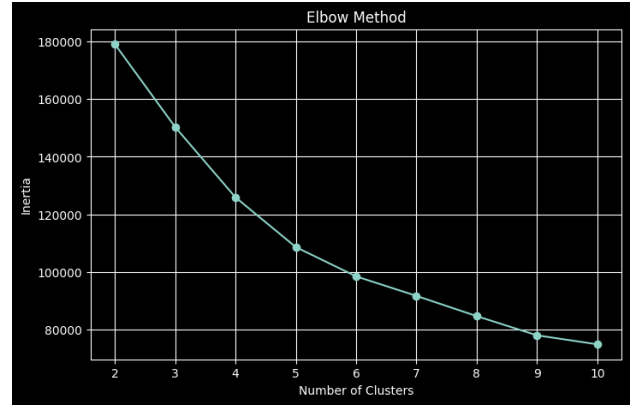


Fig. 8. Elbow Method used to determine optimal cluster count.

Cluster analysis identified:

- Highly polluted urban regions.
- Moderately polluted industrial areas.
- Relatively cleaner regions.

The clustering results were visualised in PCA space to improve interpretability.

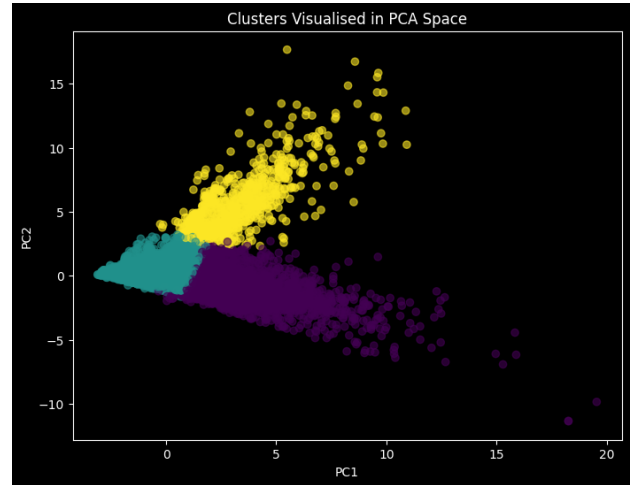


Fig. 9. K-Means clusters visualised in PCA space.

VII. CONCLUSION

This project demonstrated the successful application of exploratory data analysis techniques to an air pollution dataset from India.

The analysis identified strong relationships between pollutant variables and revealed meaningful temporal and spatial pollution patterns. Feature engineering improved the representation of pollution severity, while PCA reduced dimensionality and highlighted dominant pollutant structures.

K-Means clustering successfully grouped pollution observations into interpretable categories. Overall, the project illustrates the effectiveness of EDA techniques for understanding environmental datasets.

Future work could include additional environmental variables, advanced clustering algorithms, or predictive modelling approaches.

REFERENCES

- [1] R. Rao, "Air Quality Data in India," Kaggle Dataset.
- [2] World Health Organization, "Ambient Air Pollution," 2024.